

University of Bristol

Bristol Summer Research Internship in STEM

Bayesian Reanalysis

Project Report

[Project's Repository](#)

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1. Abstract

In psychology studies, the frequentist p-value is commonly used when evaluating the results of research. However, p-values provide only limited information as evidence for their binary threshold. To address this limitation, the Bayes factor provides an alternative means of statistical analysis in psychology due to its continuity and bidirectional measure of evidence. In this study, we computed Bayes factors in order to reanalyse 47 reproducible frequentist claims from 29 research studies published in *Teaching of Psychology* (ToP), *Psychology Learning and Teaching* (PLaT), and *Scholarship of Teaching and Learning in Psychology* (SoTL-P). As a result, 39 of 47 targets (83.0%) were concordant with the reproduced frequentist classification, 8 (17.0%) were Bayesian-inconclusive, and none were discordant. Among the 38 statistically significant targets, 33 yielded Bayesian evidence for H_1 and 5 remained inconclusive, among the 9 non-significant targets, 6 yielded evidence for H_0 and 3 remained inconclusive. Although most classifications agreed, the presence of inconclusive cases even in this optimistically selected, computationally reproducible corpus suggests that statistical significance does not necessarily imply conclusive evidential support. Through this study, we suggest that Bayes factors can serve as an effective tool for grading evidential strength in result analysis for studies. In this way, researchers can apply Bayes factors to existing results to either retrospectively reinforce concordant conclusions or highlight potential sources of inconclusive evidence that could inform improved study design.

2. Introduction

Large-scale replication projects with the objective of checking replication within psychology studies found that a substantial share of published findings could not be replicated [1]. And while full replication would require the collection of a new sample, computation reproducibility (obtaining the same result, through the same dataset) is the necessary, even if not sufficient, condition for a trustworthy result.

In addition, more often than not, part of this result is a p -value. And while a significant p validates a result, it does not account for the evidential weight: a p just below 0.05 often reflects only modest evidence, and a non-significant p is not evidence for the null but merely a failure to reject it [2]. These issues have motivated some authors to suggest an evidential reanalysis in which the support for competing hypotheses is represented explicitly [3]. Bayes factors provide one way to make that comparison.

2.1 Some Essential Bayesian Theory

This section introduces the basic Bayesian theory needed to better follow the rest of the paper, readers already comfortable with Bayes factors may skip to Section 2.2. The account follows standard textbook treatments [4, 5, 6].

2.1.1 Bayesian Inference as an Updating Machine

Bayesian inference treats an unknown quantity, this is, a parameter θ , such as an effect size not as a fixed number to be point-estimated but as something about which we are uncertain, and it represents that uncertainty with a probability distribution. Before seeing the data we summarise our uncertainty in a *prior* $\pi(\theta)$. The sampling model specifies how probable the data y would be for each candidate value of θ ; read as a function of θ for the data we actually observed, this is the *likelihood* $p(y | \theta)$. Bayes' theorem multiplies the two and renormalises,

$$\pi(\theta | y) = \frac{p(y | \theta) \pi(\theta)}{p(y)}, \quad p(y) = \int p(y | \theta) \pi(\theta) d\theta,$$

to yield the *posterior* $\pi(\theta | y)$. The interpretation is intuitive: the posterior is simply the prior re-weighted by how well each value of θ predicted the data. Values of θ that make the observed data more probable are up-weighted; values that make it less probable are down-weighted [4].

So, we have:

- $p(y | \theta)$, the *likelihood*, which describes the sampling model;
- $\pi(\theta)$, the *prior distribution*, the pre-data distribution assigned to the parameter;
- $\pi(\theta | y)$, the *posterior distribution*, our post-data uncertainty about the parameter;

- $p(y)$, the *marginal likelihood* (equivalently the *evidence*, or *prior predictive density*) of the observed data.

Because $p(y)$ does not depend on θ , it is the normalising constant that makes the posterior integrate to one (condition of it being a probability distribution), and the theorem is often written in the proportional form $\pi(\theta | y) \propto p(y | \theta) \pi(\theta)$. Yet $p(y)$ represents the average probability the model assigns to the data before they are seen, the likelihood averaged over every parameter value, weighted by the prior, and, because of that, the model's honest statement of what it expected to observe [5]. The specific default priors we place on each effect are set out later, in the Methods.

2.1.2 Comparing hypotheses: the Bayes factor

Hypothesis testing follows the same logic in Bayesian analysis. Each hypothesis H is itself a model, a likelihood together with a prior over its parameters, and so it assigns the data an evidence $p(y | H) = \int p(y | \theta, H) \pi(\theta | H) d\theta$. To weigh two hypotheses against each other we take the ratio of their evidences, the *Bayes factor*,

$$\text{BF}_{10} = \frac{p(\text{data} | H_1)}{p(\text{data} | H_0)}.$$

Applying Bayes' theorem to the hypotheses themselves shows exactly what this ratio does: the shared term $p(\text{data})$ cancels, so the relation between prior and posterior odds is clear,

$$\underbrace{\frac{p(H_1 | \text{data})}{p(H_0 | \text{data})}}_{\text{posterior odds}} = \text{BF}_{10} \times \underbrace{\frac{p(H_1)}{p(H_0)}}_{\text{prior odds}}.$$

Because of this, the Bayes factor is the factor by which the data multiply the odds of one hypothesis over the other, a direct, continuous measure of relative evidence that, unlike a p -value, can point either way and needs no rejection region to be interpreted [7].

2.1.3 Why the evidence works as an Occam's razor

The evidence $p(y | H)$ is an average of the likelihood over the prior, and this averaging gives the Bayes factor its most appealing property: it prefers the simplest hypothesis compatible with the data automatically, with no penalty added explicitly [6]. The mechanism works as a conservation law. A model's prior predictive probability must sum to one across every dataset the model could conceivably produce. This means that a very flexible model, one with many free parameters, or a broad, diffuse prior on the effect, can accommodate an enormous range of possible datasets, but for that very reason it must spread its probability thinly, assigning only a little to any particular one.

By contrast, a more constrained model concentrates its predictive mass on a narrower set of datasets; if the data we actually observe fall within that set, the constrained model assigns them a higher probability and so earns a larger evidence. MacKay [6] makes this precise by

factorising the evidence into a best-fit likelihood times an *Occam factor*. This Occams factor is no more than the ratio of the posterior width to the prior width of the parameter, which measures how sharply the data had to prune the parameter space the model began with. The consequence is that a diffuse alternative that “could have predicted almost anything” is penalised relative to a sharp null that predicts “no effect,” which is exactly why a Bayes factor can return positive evidence for H_0 when the data are consistent with it, distinguishing “the data favour no effect” from “the data are uninformative,” something no significance test can do.

2.2 Why a Bayesian Reanalysis

Wetzels et al. [8] in reanalysing 855 t tests, found that many “significant” results correspond to only anecdotal Bayesian evidence. Additionally, because both Bayesian reanalyses and revised-threshold proposals agree on the point that results near $p = 0.05$ typically carry only weak support, this motivates either using stricter thresholds or the direct reporting of Bayes factors [9, 10]. To add on to this, Etz and Vandekerckhove [11] used a Bayesian lens to reinterpret the *Reproducibility Project*, distinguishing the strength of evidence from the binary replicate/fail verdict, which led them to explain effects such as sample selection.

This project extends the logic in this existing research. However, differently, our inherited targets are not all t tests: they include factorial and mixed designs, heteroscedastic comparisons, multivariate tests, rank procedures, mixed-effects models, and latent-variable analyses. This heterogeneous reanalysis raised a methodological question that more homogeneous reanalyses can usually largely avoid: what should “the Bayesian version” of an original analysis mean when there is no clear or satisfactory one-to-one Bayesian analogue of the frequentist procedure?

Our strategy was to preserve correspondence with the reproduced analysis whenever a defensible Bayesian analogue was available, so that the Bayesian result would address as nearly as possible the same statistical question as the original test. However, when such correspondence could not be found, for example, when no credible Bayesian analogue existed, or when reproducing the test would actually distort the underlying scientific claim, then a full Bayesian evaluation of the claim was prioritised, rather than strict adherence to a specific statistical testing procedure.

2.3 Research Questions

1. How often do reproduced frequentist results and Bayesian analyses lead to the same conclusion?
2. Among statistically significant results (favouring the alternative hypothesis), how often does the Bayesian analysis provide clear evidence ($BF_{10} \geq 3$) for the alternative hypothesis, how often does it remain inconclusive ($1/3 < BF_{10} < 3$), and how often does it favour the null hypothesis ($BF_{10} \leq 1/3$)?

3. Among non-significant results (favouring the null hypothesis), how often does the Bayesian analysis provide evidence for the null hypothesis ($BF_{10} \leq 1/3$), and how often does it remain inconclusive ($1/3 < BF_{10} < 3$)?
4. Within comparable analysis types, to what extent are smaller p-values associated with the larger BF_{10} .
5. How sensitive are BF_{10} values, and the evidence classifications derived from them to reasonable alternative prior specifications?
6. Exploratorily, which statistical and design features lead to the concordance or discordance between the frequentist interpretation and Bayesian evidence classification?
7. Under what conditions does Bayesian analysis provide useful information beyond the reproduced frequentist result?

2.4 In this Study

This report follows the analysis from the reproduced frequentist results to their Bayesian re-evaluation. The **Methods** section explains how the target claims were inherited and screened, how the reproduced frequentist results were independently verified, and how each target was matched to an appropriate Bayesian analysis. It then introduces the three analysis waves, the prior specifications used for each model family, the sensitivity and computational-stability checks, and the rule used to classify frequentist–Bayesian concordance.

The **Results** begin with the overall concordance between the reproduced frequentist decisions and the Bayesian evidence classifications. We then examine how sensitive those conclusions are to alternative prior specifications and numerical approximation, before looking more closely at the claims for which the Bayesian evidence remains inconclusive. Particular attention is given to cases where the design, model structure, or available Bayesian methodology makes the comparison more informative than a simple conversion between statistical tests.

Finally, the **Discussion** considers what these results imply for the interpretation of computationally reproduced findings, when Bayesian reanalysis adds information beyond a significance decision, and what the project suggests for future Bayesian reanalyses of heterogeneous published results. Limitations of the selected corpus and of the correspondence between frequentist and Bayesian models are considered and possible extensions of the work are then outlined.

3. Methods

3.1 Claim Extraction Procedure and Scope

The data used in the present study are drawn from Ali Almuhanha and Dr Peter Allen’s (2026) study, *The Computational Reproducibility of Psychology Learning and Teaching Research* [12]. Briefly, the original study randomly sampled 60 of 103 papers identified as having open data and as potentially computationally reproducible. These 103 papers were drawn from 1,039 empirical papers published in Teaching of Psychology (ToP), Psychology Learning and Teaching (PLaT), and Scholarship of Teaching and Learning in Psychology (SoTL-P) between January 1, 2015, and May 29, 2025. For each sampled paper, the source-study researchers identified a central claim and selected its statistical result. When multiple statistical results support a same claim, the result with the largest effect size was selected as the primary target result, and an additionally secondary target was randomly selected from the other supporting statistical results.

The main unit of our analysis is a target statistical result linked to a scientific claim. Some papers contribute more than one target result, so results from the same paper will not be treated as fully independent: Descriptive summaries will be reported at the target level, alongside the number of unique papers, and this dependence will be considered when interpreting the results. To construct our dataset, the research team screened papers selected by the source study and identified target results eligible for inclusion in the present Bayesian reanalysis based on the following criteria:

1. The target result was successfully reproduced in the source study
2. A p-value was available for the target result.
3. Sufficient data or summary statistics were available to conduct the equivalent Bayesian analysis.

Composition figures of our final dataset (family counts, sample-size and role by family) are available in Appendix B. The open-data source for each study, including the correspondence between our anonymous notation to the actual study’s reference is documented in Appendix C.

3.2 Verification of the Frequentist Targets

Before any Bayesian reanalysis, each ready target was checked against the open data and the source analysis. The project recorded (i) the result reported in the article, (ii) the result reproduced by Almuhanha and Allen, and (iii) the result recomputed in the present

repository. This step was necessary because the original reproductions were run on different formats (JASP, SPSS, R), the data sources too were inherited on multiple formats (`.csv`, `.xlsx`, `.sav`) and the statistics available were approximate values.

Where the raw data and analysis specification permitted exact recomputation, we reproduced the focal statistic, degrees of freedom, and p -value in R. Small numerical differences attributable to software versions, optimisation, or numerical algorithms were documented rather than treated as substantive disagreements when they did not change the underlying test or conclusion. Five claims remain held, these represent primary (or secondary) results that were not reproducible and whose corresponding secondary (or primary) result was reproducible, as these were recomputed together.

3.3 The Three-Wave Bayesian Design

The project was designed around three waves of analysis, with each wave increasing in modelling complexity and the third wave reserved for specialised extensions. This allowed a simple-first, complex-later approach to be taken.

Wave 1 contains simple analyses that can be represented by one main effect or one correlation. These include one-sample t -tests, paired-samples t -tests with exactly two conditions or time points, equal-variance independent-groups t -tests with two groups, and Pearson correlations. A two-group ANOVA can also fall within this wave when it is exactly equivalent to the corresponding t -test, with the same sample, contrast, error term, and variance assumptions.

Wave 2 contains more complex ANOVA, regression, and multilevel analyses. This includes one-way ANOVAs with more than two groups, factorial ANOVAs, interactions, models allowing unequal group variances, repeated-measures ANOVAs with more than two conditions or time points, within-subject factorial designs, mixed designs, and multilevel models.

Wave 3 contains specialised multivariate, structural-equation, rank-based, ordinal, longitudinal, and other non-standard analyses.

3.4 Correspondence-first reanalysis protocol

The central methodological rule we wanted to follow was to preserve the reproduced statistical hypothesis whenever a defensible Bayesian correspondence existed. However, when a satisfactory correspondence could not be established, a direct Bayesian model of the inherited scientific claim was constructed.

Additionally, in the Bayesian reanalyses, sidedness always followed the reproduced statistical test, with the exception of the claim-level reconstructions, which followed the inherited scientific claim.

3.5 Bayesian Models and Prior Specification

Every Bayes factor depends on both the likelihood and the prior specification. Our guiding principle was to use an established default Bayes-factor prior where one existed, and an explic-

itly justified weakly informative, fractional, or encompassing prior where no suitable standard default was available. Table 3.1 lists the primary prior and the sensitivity specifications used for each family.

Family (studies)	Primary prior	Primary scale ¹	Scales computed
Wave 1: standard/default Bayes-factor families			
t tests	Cauchy(0, r) on δ (JZS) [13]	$r = 0.707$	0.5 / 0.707 / 1.0
Correlation	Stretched beta on ρ [14]	$r = 1$ (uniform)	0.5 / 1.0 / 1.5
Wave 2: ANOVA, regression, and multilevel models			
Factorial / mixed ANOVA	Multivariate Cauchy on fixed effects [15]	$r_{\text{fixed}} = 0.5$	0.25 / 0.5 / 0.707
Linear regression	Mixture of g priors [16, 15]	$r = 0.5$	0.354 / 0.5 / 0.707
Model-averaged t (13)	Cauchy on δ , with evidence averaged over equal- and unequal-variance Normal models [13, 17]	$r = 0.707$	0.5 / 0.707 / 1.0
Adjusted contrast (43)	Normal prior on the focal standardized contrast; bridge-sampled marginal likelihood [18, 19]	SD = 0.6	0.3 / 0.6 / 1.0
Heteroscedastic ANOVA (55)	Normal prior on the focal contrast with group-specific residual variances [18, 19]	SD = 0.5	0.25 / 0.5 / 0.707
Logistic GLMM (37)	Normal prior on the focal log-odds coefficient [18, 19]	SD = 0.6	0.3 / 0.6 / 1.0
Wave 3: specialised and claim-level models			
MANOVA (03, 18, 44)	Adjusted fractional Bayes factor with a model-specific minimal training fraction [20]	—	primary only
SEM order restriction (20)	Adjusted fractional/order-restricted Bayes factor for $\psi > 0$ [21, 22]	baseline fraction $\times 1$	$\times 1 / \times 2 / \times 3$
Signed-rank (26)	Cauchy prior on latent δ [23]	$r = 0.707$	0.5 / 0.707 / 1.0
Spearman ρ (39)	Stretched beta prior on latent ρ [23]	$\kappa = 1$	0.5 / 1 / 2
Longitudinal claim model (40)	Proper Normal priors on focal post-intervention deviations δ_4, δ_5 ; encompassing-prior order Bayes factor [18, 24]	SD = 0.5	0.25 / 0.5 / 1.0
Ordinal claim model (45)	Cumulative-probit mixed model with orthonormal dimension contrasts and symmetric Normal priors on contrast coefficients [25, 24]	SD = 1	primary ordinal specification

Table 3.1: Primary priors and prior-scale specifications used in the Bayesian reanalysis.

Wave 1 For the t tests we used the Jeffreys–Zellner–Siow (JZS) prior on the standardized mean difference δ : a Cauchy(0, r) distribution, characterised by being heavy-tailed (specially when compared to the Normal distribution), which is relevant here so that the Bayes factor is not immediately more strongly dismissive of extreme values. The primary scale $r = \sqrt{2}/2 \approx 0.707$ is the R package `BayesFactor` “medium” default, while $r = 0.5$ and $r = 1$ are the “narrow” and “wide” alternatives, respectively, for sensitivity analysis [13, 14, 26]. For correlations we use the stretched-beta prior on ρ ; the primary scale $r = 1$ makes this the

¹The methodological citations identify the main source for the corresponding Bayesian prior, model class, or Bayes-factor construction. For custom model-based analyses, the exact narrow, primary, and wide scales were selected for the present reanalysis rather than prescribed by those sources.

uniform prior on $(-1, 1)$, with $r = 0.5$ and $r = 1.5$ as the narrower and wider alternatives [14].

Wave 2. Factorial and mixed ANOVAs use the multivariate Cauchy construction of Rouder et al. [15] on standardized fixed effects, with primary fixed-effect scale $r_{\text{fixed}} = 0.5$. Regressions use the mixture-of- g -priors extension [16] with primary $r = 0.5$. The unequal-variance t analysis in Study 13 uses model averaging over equal- and unequal-variance normal models while retaining the Cauchy prior on δ [17]. The heteroscedastic one-way design in Study 55 uses a Normal focal-contrast prior with separate per-cell variances. Study 37 uses a weakly informative $\text{Normal}(0, 0.6^2)$ prior on the focal log-odds coefficient, with bridge sampling used for model comparison [27, 18]. The 0.6 scale is a project-calibrated choice; Gelman’s broader logistic recommendations are used only as a secondary comparator rather than as the justification for the primary prior [28].

Wave 3 The MANOVA targets use adjusted fractional Bayes factors. The relevant training fraction is determined by the fractional-Bayes-factor construction and model dimensions; it should not be interpreted as one universal $b = J/n$ rule applying to every multivariate design [29, 21, 20, 30]. Study 20 tests the inherited one-sided SEM contrast $\psi > 0$ using an order-restricted adjusted fractional comparison. Studies 26 and 39 use project-adapted latent-normal rank-likelihood implementations motivated by the Bayesian rank-testing literature [23]; these avoid modelling the observed ranks as Gaussian measurements, but they still rely on a latent-normal/correlation structure and are not literally assumption-free.

Studies 40 and 45 are the two claim-level reconstructions. Study 40 models the longitudinal exam scores directly and defines focal post-intervention deviations δ_4 and δ_5 . The inherited claim is

$$H_{40} : \delta_4 > 0 \ \& \ \delta_5 > 0.$$

The focal priors are independent symmetric Normal distributions, with primary SD = 0.5 and sensitivity scales 0.25 and 1.0. Under the symmetric encompassing prior, $P(H_{40}) = 1/4$, and the Bayes factor against the complement is obtained by dividing posterior odds by prior odds.

Study 45 models the raw 1–7 ratings using a cumulative-probit mixed model with participant and attribute random intercepts. The four dimensions are represented with orthonormal contrasts and a symmetric $\text{Normal}(0, 1)$ prior on the three contrast coefficients so that no dimension is privileged by the parameterisation. The focal hypothesis is

$$H_{45} : \alpha_{\text{Import}} > \max(\alpha_{\text{Aware}}, \alpha_{\text{Explain}}, \alpha_{\text{Dev}}) \ \& \ \alpha_{\text{Dev}} < \min(\alpha_{\text{Import}}, \alpha_{\text{Aware}}, \alpha_{\text{Explain}}).$$

Under the symmetric prior, two of the $4! = 24$ strict orderings satisfy this hypothesis, giving $P(H_{45}) = 1/12$. The order-restricted Bayes factor is therefore again computed from posterior odds divided by prior odds. The ordinal model is the primary claim-level analysis; a Gaussian mixed model on dimension averages is retained only as a secondary cross-model validation.

3.6 Software and Reproducible Analysis Pipeline

All analyses are implemented in R and organised in a version-controlled repository containing the prepared open data, claim registry, analysis scripts, configuration files, generated tables, and figures. Package versions are locked with `renv.lock` so that the computational environment can be reconstructed².

The project uses packages including `BayesFactor`, `BFpack` [30], `bain`, `RoBTT`, `brms` [18], `bridgesampling` [27], and `lavaan` [31]. The complete package environment, including exact package versions and computational dependencies, is recorded in the [repository](#). [32]

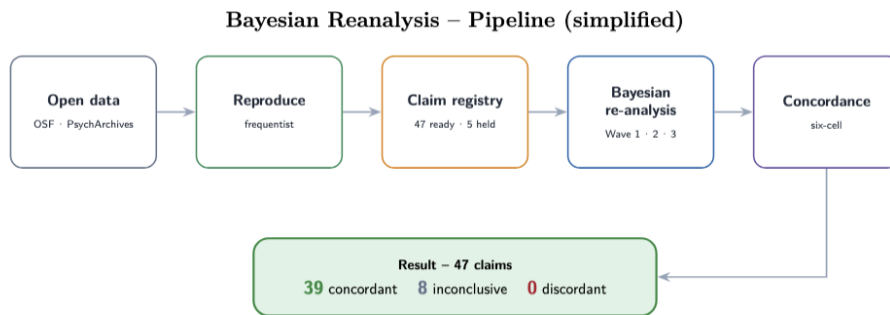


Figure 3.1: A simplified scheme of the pipeline (full version can be found in Appendix D)

A single `00_run_all.R` driver runs frequentist recomputation, then the three Bayesian waves, then classification and figures; MCMC analyses are seeded so reruns reproduce the reported Bayes factors, but stability was tested and reported for each case.

Priors, claim readiness, and data provenance live in `config/`, so analytic choices can all be inspected in one place.

3.7 Concordance Rule

The concordance rule used was the following: On the frequentist side, significant ($p < 0.05$) versus non-significant ($p \geq 0.05$), using the reproduced p at the sidedness the source reported; On the Bayesian side, H_1 ($BF_{10} \geq 3$), inconclusive ($1/3 < BF_{10} < 3$), or H_0 ($BF_{10} \leq 1/3$), when evaluated at the primary prior.

These rules give a six-cell concordance output (Table 4.1), where:

- Concordant: significant $\times H_1$, or non-significant $\times H_0$
- Bayes-inconclusive = either significance verdict $\times$ inconclusive;
- Discordant = significant $\times H_0$, or non-significant $\times H_1$.

The 3 and 1/3 thresholds follow the conventional Jeffreys boundary between anecdotal and substantial evidence [33, 34].

²[See all packages and dependencies].

BF_{10}	Interpretation
≥ 100	Decisive evidence for H_1
30 to < 100	Very strong evidence for H_1
10 to < 30	Strong evidence for H_1
3 to < 10	Substantial evidence for H_1
1 to < 3	Anecdotal evidence for H_1
$= 1$	Equal support for H_0 and H_1
$> 1/3$ to < 1	Anecdotal evidence for H_0
$> 1/10$ to $\leq 1/3$	Substantial evidence for H_0
$> 1/30$ to $\leq 1/10$	Strong evidence for H_0
$> 1/100$ to $\leq 1/30$	Very strong evidence for H_0
$\leq 1/100$	Decisive evidence for H_0

Table 3.2: Interpretive categories for BF_{10} , adapted from Wetzels et al. [8], based on Jeffreys (1961)[33].

3.8 Sensitivity and Robustness Analyses

Sensitivity and robustness are separated into three questions: sensitivity to the prior, numerical stability of the computation, and adequacy of the chosen likelihood/model.

3.8.1 Prior-scale sensitivity

Each family, where it had a substantively meaningful free prior scale, was evaluated at narrow/primary/wide prior scales; the per-claim span of $\log_{10} BF_{10}$ and any change in the evidence class were recorded [35]. If sensitivity to scale was severe, the cases were evaluated individually with further analyses.

Fractional Bayes-factor families are handled according to their own construction. They are not forced into a fictitious “narrow/medium/wide” effect-scale analysis when the relevant sensitivity parameter is the training fraction or when the method supplies a minimal-information fraction.

3.8.2 Computational stability

For MCMC-based Bayes factors (GLMM, bridge sampling, `blavaan`) convergence and Monte-Carlo stability were always checked. Whenever we found an estimate that was numerically noisy but overwhelmingly concordant the result was retained, because the classification is invariant to that noise. This sort of instability is expected precisely with extremely large Bayes factors and does not often affect smaller ones, which include any borderline (inconclusive) case, where more careful consideration is evidently required.

3.8.3 Secondary analyses

Several claims received additional analyses because they addressed a specific methodological concern rather than simply adding another prior. Study 37 was recomputed under a broad Cauchy prior as an alternative prior comparison. Study 55 was additionally analysed

using an adjusted-fractional Bayesian Welch approach. Study 40 was fitted with a differential pre-intervention trend so that δ_4 and δ_5 represented post-intervention deviations beyond both the baseline group gap and an extrapolated group-specific pre-trend. Study 45 was also analysed with a Gaussian mixed model on dimension averages as a cross-model validation of the primary ordinal result.

Study / family	Secondary analysis	Purpose
Logistic GLMM (37)	Cauchy(0, 2.5) focal prior, bridge sampling	Alternative-prior comparison
Heteroscedastic ANOVA (55)	Adjusted-fractional Bayesian Welch analysis	Objective/fractional comparison to the registered Normal-contrast BF
Longitudinal claim model (40)	Differential pre-trend model	Test whether the positive post-intervention deviations survive allowance for group-specific pre-intervention change
Ordinal claim model (45)	Gaussian mixed model on dimension averages	Cross-model validation of the dimension ordering

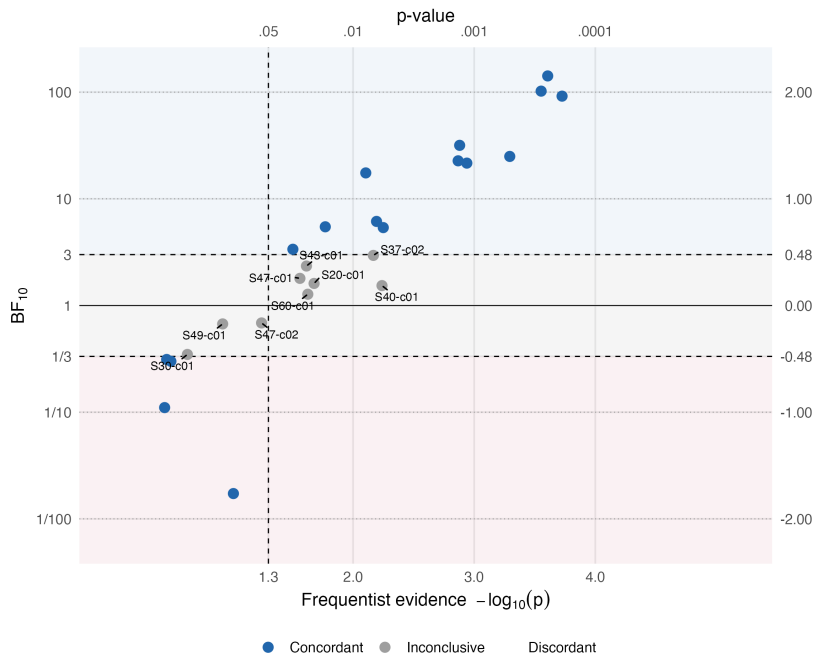
Table 3.3: Secondary and robustness analyses beyond the primary prior specifications. These analyses answer specific robustness questions and do not replace the registered primary model merely because they produce a different numerical Bayes factor.

4. Results

A total of 47 targets were included in the analysis. Unless otherwise stated, the Bayesian classification results correspond to the pre-specified primary prior for each model family. A detailed results table can be found [\[here\]](#) and the detailed concordance table can be found [\[here\]](#).

Frequentist–Bayesian Evidence Plane

Left axis = BF_{10} on the Jeffreys scale; dashed = decision thresholds; zoomed to the ± 100 decision zone



Full spread

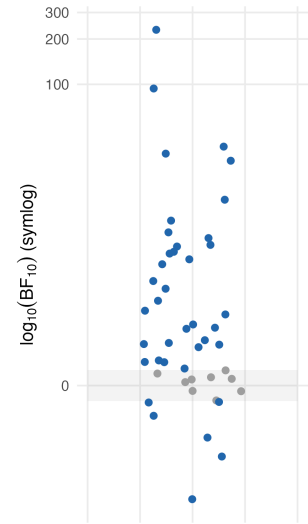


Figure 4.1: Evidence Plane

4.1 Overall Concordance

There was complete agreement in evidential direction. Of the 47 targets, 39 (83.0%) were concordant under the operational classification rule, 8 (17.0%) were Bayes-inconclusive, and none were discordant (Figure 4.2).

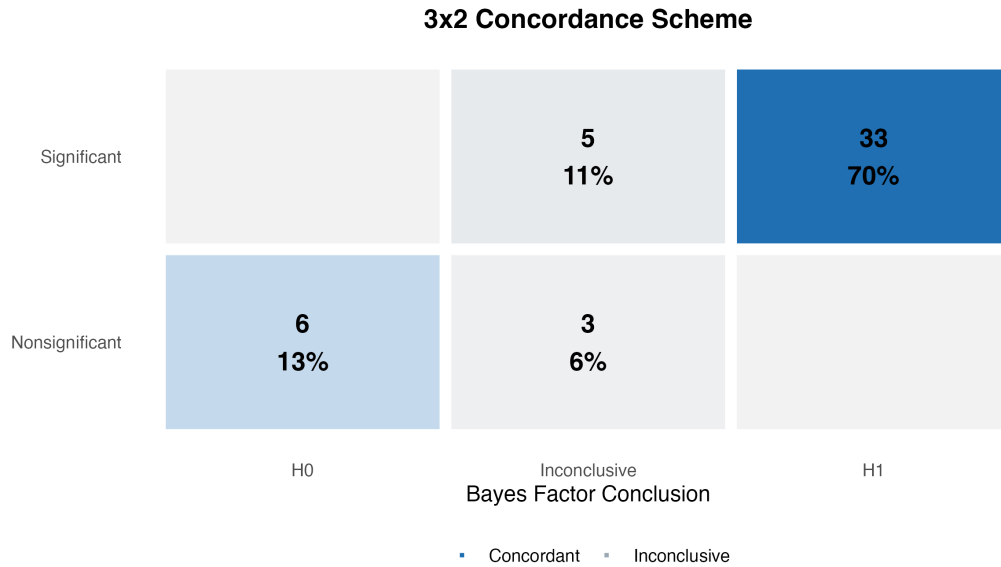


Figure 4.2: Concordance classification

Thus, no statistically significant result produced Bayesian evidence for H_0 , and no non-significant result produced Bayesian evidence for H_1 . Which means that all departures from concordance reflected insufficient Bayesian evidence in either direction and not support for the opposite hypothesis.

Additionally, among the 38 statistically significant targets, 33 (86.8%) yielded evidence for H_1 and 5 (13.2%) remained inconclusive. Among the 9 non-significant targets, 6 (66.7%) produced evidence for H_0 and 3 (33.3%) remained inconclusive. Hence, Bayesian analysis distinguished non-significant results that positively supported the null from results that remained insufficiently informative in either direction.

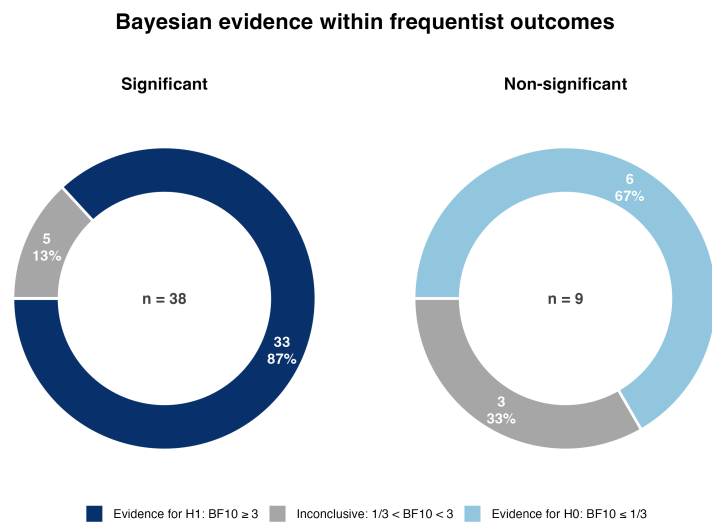


Figure 4.3: Distribution of Bayes Factor within Frequentist Results

4.2 Prior Sensitivity and Classification Stability

Forty-one of the 47 targets were evaluated under multiple prior or fractional specifications in the current sensitivity registry. The remaining targets belong to model families for which the primary analysis does not yet have a comparable narrow/primary/wide effect-scale sweep.

The classification stability was analysed on the $BF_{10} = 3$ or $BF_{10} = 1/3$ thresholds. Seven of the 41 evaluated claims (17.1%) crossed that classification boundary. Importantly, none moved directly from evidence for H_1 to evidence for H_0 , or vice versa. Amongst these seven were studies 30, 37 Claim 2, 39 Claim 2, 43 Claim 1, 49 Claim 2, and 51 Claim 2, these will be analysed in more detail below.

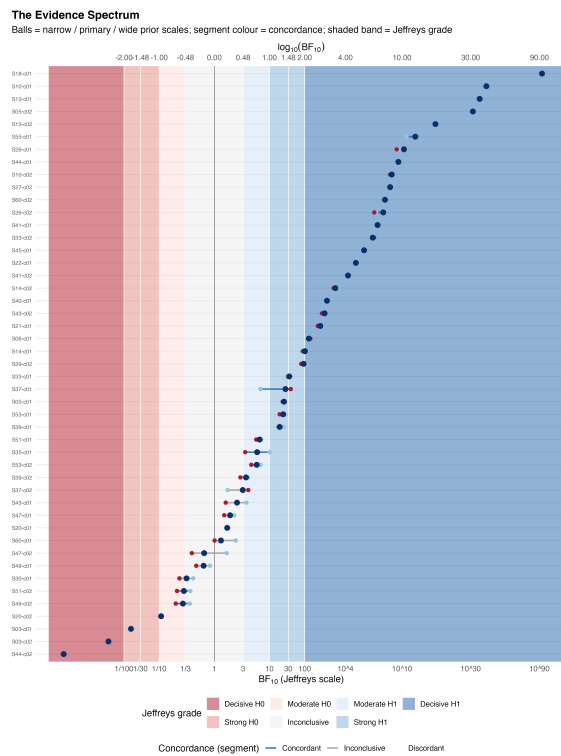


Figure 4.4: Scale sensitivity plot

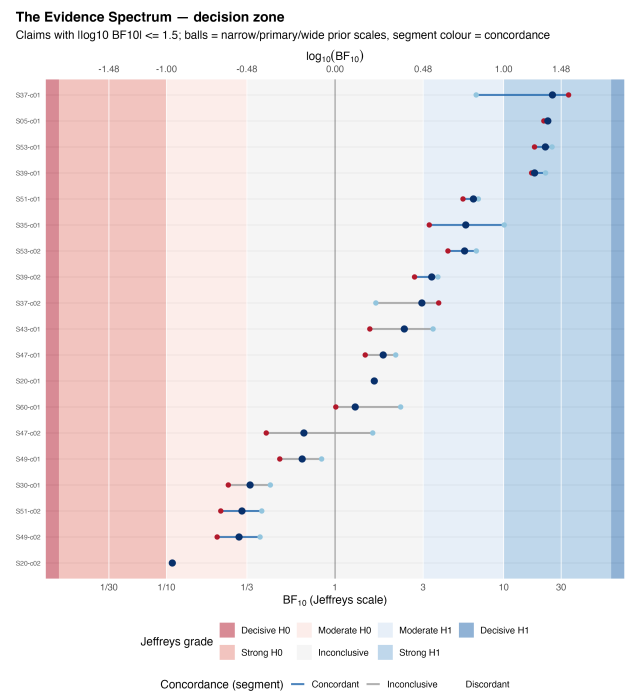


Figure 4.5: Scale sensitivity near threshold

4.2.1 Inconclusive cases

Claim	p -value	Primary BF_{10}	Sensitivity range	Interpretation
Significant				
S20C01	$p = 0.021$	1.61	1.61–2.79	Consistently inconclusive
S37C02	$p = 0.007$	2.95	1.64–3.70	Near H_1 threshold; classification-sensitive
S43C01	$p = 0.024$	2.34	1.52–3.43	Near H_1 threshold; classification-sensitive
S47C01	$p = 0.027$	1.80	1.44–2.10	Consistently inconclusive
S60C01	$p = 0.024$	1.28	1.01–2.24	Consistently inconclusive
Non-significant				
S47C02	$p = 0.057$	0.69	0.43–1.58	Consistently inconclusive
S49C01	$p = 0.119$	0.67	0.51–0.85	Consistently inconclusive
S30C01	$p = 0.233$	0.35	0.26–0.45	Near H_0 threshold; classification-sensitive

Table 4.1: Structure of the eight Bayesian-inconclusive cases. For more details see Appendix A

A relevant result of the sensitivity analysis is that the few classification changes were local. S37C02 and S43C01 crossed the conventional $BF_{10} = 3$ boundary under one prior specification, while S30C01 crossed the reciprocal $BF_{10} = 1/3$ boundary. However, all three primary Bayes factors already lay close to those thresholds and the sensitivity ranges remained close to the same evidential region, with the maximum BF_{10} value reaching only 3.7 in S37C2¹. Additionally, in the cases where the threshold was crossed, it was only by one of the prior's scar, with each result having at least two scales within the inconclusive area. Hence, in no case can we say that there has been a substantive reversal in the interpretation of the evidence.

4.2.2 Boundary cases

Claim	p -value	Primary BF_{10}	Sensitivity range	Evidential interpretation
Significant				
S39C02	$p = 0.031$	3.37	2.68–3.66	Near H_1 threshold; classification-sensitive
Non-significant				
S49C02	$p = 0.319$	0.30	0.22–0.40	Near H_0 threshold; classification-sensitive
S51C02	$p = 0.346$	0.31	0.23–0.41	Near H_0 threshold; classification-sensitive

Table 4.2: Boundary cases in which the primary Bayes-factor classification is concordant but at least one sensitivity specification falls within the inconclusive region.

Three additional claims were sensitive to the conventional evidence boundary used, but classified as concordant under the primary prior. In each case, however, the change was local: S39C02 moved from modest evidence for H_1 to an inconclusive value just below $BF_{10} = 3$,

¹See 4.5.1 for a more detailed breakdown of Study 37

while S49C02 and S51C02 moved from modest evidence for H_0 to inconclusive values just above $BF_{10} = 1/3$.

These cases reinforce the importance of looking at the Bayes factors as a continuous measure of evidence. For example, let's put two cases in perspective. Study 39 Claim 2 produced Bayes factors ranging from 2.68 to 3.66 across the prior specifications, meaning that the observed data were approximately 2.7 to 3.7 times more likely under H_1 than under H_0 . Study 37 Claim 2 ranged from $BF_{10} = 1.64$ to 3.70, so the data were approximately 1.6 to 3.7 times more likely under H_1 than under H_0 . So, for both cases, the data are only about three times more likely under H_1 than under H_0 , interpretation is stable but the concordance is different. Under a frequentist significant versus non-significant interpretation these cases make a very big difference and leave space for situations like p -hacking, under the Bayesian analysis our interpretation of the result clearly shows that the data are not strongly discriminative between our hypothesis and whatever conclusion we'd make would have to be a careful one.

4.3 Numerical and Algorithmic Stability

Because several Bayes factors were estimated using Monte Carlo, MCMC, or bridge-sampling procedures, numerical stability was assessed separately from prior sensitivity and found non-problematic: there were no significant interpretation changes to the results that were caused directly by algorithmic or numerical approximation considerations. Detailed results can be found in Appendix E

4.4 Claim-level Reconstructions: Studies 40 and 45

4.4.1 Study 40

In study 40 the frequentist target was a chi-square difference test, $\Delta\chi^2(2) = 10.31$, $p = .006$. The final Bayesian analysis didn't attempt to reproduce that model-comparison statistic directly. Instead, it modelled the claim that the opt-in group showed additional positive post-intervention change. Under the primary focal prior, the estimated deviations were $\delta_4 = 0.245$ and $\delta_5 = 0.224$, with posterior probability

$$P(\delta_4 > 0, \delta_5 > 0 \mid y) = 0.996.$$

The H_1 classification remained unchanged across the focal prior sweep $SD = 0.25, 0.5, 1.0$.

As a secondary robustness check, a differential pre-intervention trend was added so that δ_4 and δ_5 represented additional post-intervention deviations beyond both the baseline group difference and an extrapolated group-specific pre-trend. Under this more demanding model, the posterior ordering probability decreased to 0.888, but the Bayes factor remained $BF_{10} = 23.76$. So, the evidence was attenuated by the pre-trend adjustment but remained clearly in favour of what was claimed. Consequently, Study 40 was classified as significant $\times$ Bayesian

H_1 .

4.4.2 Study 45

For Study 45 the frequentist target was a highly significant Friedman test. Because no satisfactory direct omnibus Bayesian Friedman analogue was found, the final analysis modelled the substantive ordering on the raw 1–7 item responses using a cumulative-probit mixed model with student and attribute random intercepts. The fitted dimension offsets reproduce the substantive ordering found in the Gaussian validation model: **Importance** is highest and **Development** lowest.

The final symmetric-prior ordinal fit had maximum $\hat{R} = 1.0021$, minimum ESS = 1994, and zero divergences. The empirical prior probability of the focal order restriction was 0.082365, close to the theoretical 1/12, while all retained posterior draws satisfied the ordering. The encompassing-prior calculation gave $BF_{10} \approx 2.67 \times 10^5$, decisively favouring the dimension-ordering that was claimed by the study. A Gaussian mixed model on dimension averages gave the same ordering and a comparably decisive encompassing comparison, providing cross-model validation. Consequently, Study 45 was classified as significant $\times$ Bayesian H_1 .

4.5 What drives disagreement

The clearest driver of Bayes-inconclusive evidence is the strength of the frequentist signal. All 27 statistically significant targets with $p < 0.005$ produced Bayesian evidence for H_1 , while all targets with p -values between 0.021 and 0.027 were Bayesian-inconclusive. However, this pattern was not absolute, one claim (Study 37 claim 2) remained inconclusive despite $p < 0.01$, indicating that p -value magnitude alone does not determine the Bayes-factor classification, this case will be better analysed in 4.5.1. Nevertheless, we can say that being near the frequentist threshold boundary was the primary indicator of Bayesian anecdotal evidence, rather than statistical family or sample size.

Reproduced p -value	n	Bayesian H_1	Inconclusive	Inconclusive (%)
$p < .005$	27	27	0	0.0
$.005 \leq p < .01$	5	4	1	20.0
$.01 \leq p < .02$	1	1	0	0.0
$.02 \leq p < .03$	4	0	4	100.0
$.03 \leq p < .05$	1	1	0	0.0

Table 4.3: Bayesian evidence classification of the 38 statistically significant targets by reproduced p -value band.

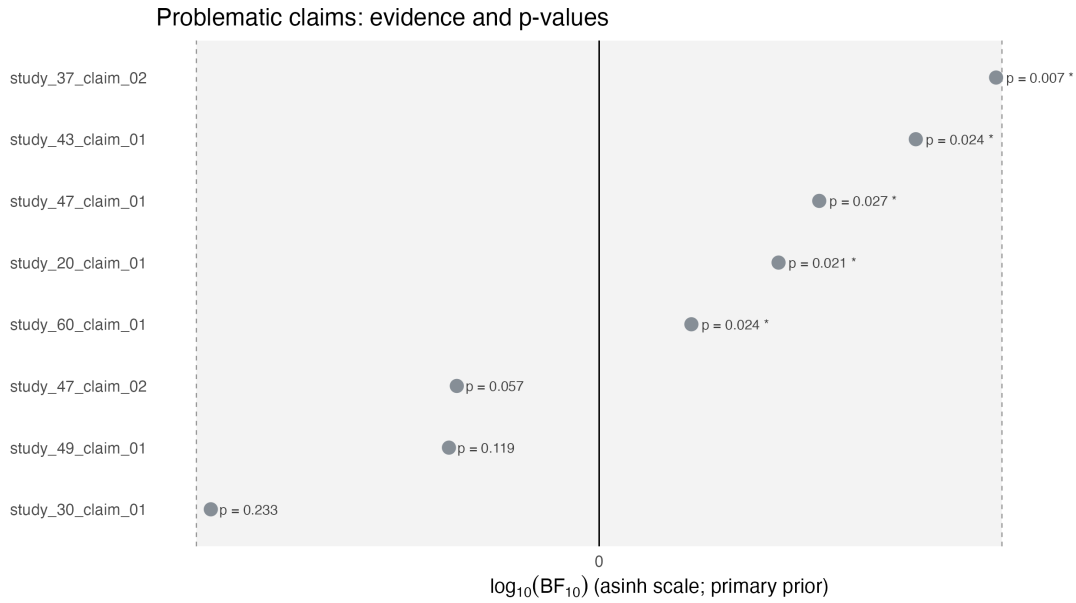


Figure 4.6: Bayes factor positioning versus p -value per claim classified as inconclusive (boundaries represent $BF_{10} = 3$ or $1/3$ threshold)

Additionally, in this corpus, on the significant side, small samples did not, in general, provide a general explanation for inconclusive results. Significant-inconclusive targets had a median $N = 153$, compared with $N = 156$ among significant-concordant targets.

Result group	n	Median p	Median N	Median BF_{10}
Significant + H_1	33	1.22×10^{-6}	156	14,725
Significant + inconclusive	5	.0236	153	1.80
Non-significant + H_0	6	.353	161.5	0.064
Non-significant + inconclusive	3	.119	37	0.672

Table 4.4: Characteristics of results summary

Interestingly, the mechanism differs on the non-significant side. Non-significant-inconclusive results have a median of $N = 37$, compared to $N = 161.5$ for non-significant results that produce Bayesian evidence for H_0 . Here Bayesian analysis clearly exposes a case of insufficient information, where a sample can fail to reject H_0 without containing enough information to favour it positively. This is the clearest demonstration in the corpus of the difference between *evidence of absence* and *absence of evidence* [36, 2]: with enough information a non-significant result can positively support the null, whereas a small sample can fail to reject the null without containing enough information to support it. So in this reanalysis we could clearly distinguish, amongst the non-significant cases, the ones where the data favoured H_0 from the ones where it just couldn't discriminate.

4.5.1 Case study: Study 37, Claim 2

Study 37 compared asynchronous versus synchronous online courses on student withdrawal rates in Introductory Psychology during Covid-19. It was modelled through a full Bayesian Generalized Linear Mixed Model (GLMM), using bridge-sample. The primary prior was the GLMM full with focal $SD = 0.6$, giving a Bayes factor of 2.949 for claim 2.

The registered scale the Bayes factor value varied with BF_{10} narrow ($r = 0.3$) = 1.641, BF_{10} primary ($r = 0.6$) = 2.949, BF_{10} wide ($r = 1$) = 3.704. So the evidence strength is sensitive to the prior scale, although, only the wide prior reaches substantial evidence ($BF_{10} > 3$) and remains quite near the threshold. As a further robustness check, Gelman’s generic Cauchy(0, 2.5) prior on the focal coefficient (Gelman et al. [28]) was computed and gave a $BF_{10} = 1.85$, which reiterated the inconclusive interpretation, giving and even more conservative value than the project’s Normal(0, 0.6).

	Claim 1: family obligations	Claim 2: synchronous format
Frequentist estimate, $\hat{\beta}$	1.2265	-1.1135
Frequentist SE	0.3528	0.4114
Wald z	3.476	-2.706
p	0.000509	0.006801
N	563	563
Primary BF_{10}	24.96	2.95
Bayesian classification	H_1	Inconclusive

Table 4.5: Comparison of the two focal claims in Study 37. Both claims were estimated from the same sample, but Claim 2 produced weaker frequentist and Bayesian evidence despite a similar coefficient magnitude.

This result is especially interesting when contrasted with the study’s first claim, which had conclusive Bayesian evidence for H_1 with a $BF_{10} = 24.96$. The difference cannot be explained by sample size, since both claims use the same $N = 563$, nor simply by coefficient magnitude, since the reproduced frequentist coefficients were similar in absolute size (1.2265 vs -1.1135). A relevant factor is that the frequentist GLMM estimated claim’s 2 coefficient less precisely than the family-obligations coefficient ($SE = 0.411$ vs. $SE = 0.353$). Although this seems like a small difference, it should not be considered in isolation. Claim 2 also has a slightly smaller absolute coefficient, so the combined standardized signal falls from

$$|z_1| = 3.48 \quad \text{to} \quad |z_2| = 2.71.$$

Under the local quadratic approximation to a regular likelihood, the improvement in log likelihood associated with a single coefficient is approximately $z^2/2$. So the difference between the two claims is approximately

$$\frac{3.48^2 - 2.71^2}{2} \approx 2.38$$

log-likelihood units, corresponding to a likelihood-scale ratio of

$$\exp(2.38) \approx 10.8.$$

Thus, what appears to be a relatively small difference in coefficient precision translates into a much larger difference in the strength of the likelihood signal.

There is also a design-level explanation for this loss of precision. Although both claims use the same $N = 563$ students, family obligations varies between individual students, whereas synchronous course format varies between course sections. With students clustered within only 10 sections, the nominal sample size does not represent the same amount of independent information for the section-level course-format effect. In this sense, the larger standard error for Claim 2 can be understood as a consequence of weaker information about that coefficient.

The calculation above is explanatory rather than the computation of the Bayes factor. The reported BF_{10} values come from bridge-sampled marginal likelihoods for the complete GLMMs, which average over the focal prior, nuisance parameters, and section-level random effect. Nevertheless, the observed Bayes-factor ratio,

$$\frac{24.96}{2.95} \approx 8.46,$$

is of the same order as the difference anticipated from the local likelihood geometry.

Additionally, this comparison illustrates an important distinction between frequentist and Bayesian evidence. Both approaches regard Claim 1 as more strongly supported than Claim 2: the Wald statistic falls from 3.48 to 2.71 and the p -value increases from 0.0005 to 0.0068, while BF_{10} falls from 24.96 to 2.95. The difference really lies in what is being quantified. The frequentist $p = 0.0068$ for Claim 2 measures how unusual the observed result would be under the null model, whereas the Bayes factor asks whether the observed data were better predicted by the full model than by the reduced model after averaging over the prior distribution of its parameters. On that comparison the evidence for Claim 2 is positive but only in about a 3:1 ratio.

5. Discussion

5.1 Principal findings

The project can report two related findings: (1) the empirical finding about the evidential status of the frequentist corpus when reanalysed with Bayesian methods (2) a methodological finding about how heterogeneous Bayesian reanalyses can be conducted.

Firstly, across 47 target results, 39 were classified as concordant, 8 as Bayes-inconclusive and none as discordant. This does not entail that Bayesian and frequentist evidence always agree in direction, but rather that in this corpus that did not occur. The main result is that in 17% of cases Bayesian analysis changed how confident we were in some results. In addition, specific drivers were hypothesised for the change in evidential strength, with a near-threshold p -value being the main one, but also precision in estimates more than sample size as a determinant. On top of that, the usefulness of Bayesian analysis for making the distinction "absence of evidence is not evidence of absence" proved itself a valuable result.

Secondly, methodologically, the project shows that "Bayesian reanalysis" is not always a straightforward correspondence and that there are model types to which that correspondence is more or less direct. The correspondence-first protocol helps keep the comparison more honest to the original analysis while not forcing experimental or unclear Bayesian models usage.

5.2 Note on an optimistic sample

The high concordance rate should be interpreted in light of how the sample was selected. The project only includes open-data studies that could be reproduced and reanalysed, and when several results supported the same claim, the largest-effect result was selected as the primary target. These choices favour clearer and stronger frequentist results, so the observed 83.0% concordance will almost certainly be an overestimate.

The sample is also heavily weighted toward statistically significant results. Of the 47 targets, 38 were significant and only 9 were non-significant. The influence of publication bias is another argument that can be made for the Bayesian evidence, where we actually find that the data support an alternative hypothesis to the one postulated.

5.3 When does a Bayesian reanalysis add value?

The reanalysis adds most information in three situations: (i) It distinguishes threshold crossing from strong evidence. A reproduced result just below $p = 0.05$ can be statistically

significant while the competing Bayesian models remain difficult to distinguish. (ii) It distinguished failure to reject from positive evidence for the null. (iii) Complex analyses force assumptions that are easy to hide in a single frequentist test statistic: prior scale, parameterisation, dependence structure, and the exact hypothesis being compared. Hence, sensitivity analysis can reveal when an evidential statement depends strongly on choices that deserve substantive justification.

The reanalysis added less when the data already discriminate overwhelmingly between the relevant hypotheses. In such cases the Bayes factor mainly quantified a conclusion that was already difficult to doubt.

Studies 40 and 45 add a fourth methodological lesson. The most challenging part of a reanalysis can be when an exact cross-framework translation cannot be found or properly argued for. In that situation we found that explicit claim-level reconstruction was preferable to presenting a project-specific analogue.

5.4 Recommendations

For those aiming to follow a Bayesian reanalysis with a heterogenous corpus such as this one, we recommend:

1. **Reproducing before reinterpreting.** Reproducing the published result seemed to be the first step of the reanalysis, once it guaranteed that it wasn't built on an unresolved computational mismatch. In fact, if a frequentist result could not be recomputed from the same data that would then be used in the reanalysis, then no meaningful comparison could be made.
2. **Registering the claim–result trace.** Recording exactly which scientific claim, statistical result, model design, estimand, contrast, sidedness, and data structure are being reanalysed.
3. **Keeping a registry of exact results.** Recording the main results and variable values used up to machine precision can avoid rounding doubts and allow for accurate analyses without the need to access the raw data.
4. **Using correspondence first.** Where a well-established Bayesian analogue exists, preserve the reproduced statistical hypothesis as closely as possible rather than redesigning it from the prose of the article. If no satisfactory analogue can be defended, then model the inherited scientific claim directly, always documenting why correspondence failed.
5. **Making priors part of the reported hypothesis.** Distinguish literature defaults from project choice and report the primary prior explicitly, and perform sensitivity analysis when the evidential conclusion could plausibly depend on the prior.
6. **Separating prior sensitivity, numerical stability, and model adequacy.** They are different failure modes and correspond to different diagnostics.

7. **Reporting continuous evidence as well as categories.** The 1/3 and 3 boundaries are useful summaries, not natural discontinuities in evidence.

Additionally, our results suggest several practical ways of reducing the probability that a conventionally significant frequentist result nevertheless produces inconclusive Bayesian evidence. These recommendations cannot guarantee a conclusive Bayes factor, since the realised evidence depends on the observed data, model and prior, but they can make an inconclusive outcome substantially less likely.

1. **Being cautious with p -values close to the significance threshold.** A result just below $p = .05$ may be statistically significant while still being based on fairly weak evidence. In our reanalysis, almost all cases with a p -value near the threshold were classified as Bayesian-inconclusive.
2. **Aiming for precise estimates, not only for statistical significance.** When planning a study, it is useful to think not only about whether the effect is likely to reach $p < .05$, but also about how precisely the focal effect will be estimated. Smaller standard errors and narrower confidence intervals make the result more informative. This can be improved through better measurement, balanced designs, repeated observations where appropriate, and more efficient use of the available data.
3. **Making sure there is enough information at the level where the focal predictor varies.** A large total N does not necessarily provide a large amount of independent information about every effect. In Study 37, for example, both claims involved $N = 563$ students, but synchronous course format varied across only 10 sections. So, information about that effect depended heavily on comparisons between sections rather than simply on the total number of students. For cluster-level predictors, increasing the number of independent clusters can be more useful than adding participants within existing clusters units.
4. **Making non-significant results capable of supporting a substantive conclusion.** A non-significant result can arise either because the effect is genuinely small or because the study is too imprecise to detect it. In our corpus, six of the nine non-significant results supported H_0 , while three remained inconclusive, illustrating that non-significance alone does not always provide a clear answer.

5.5 Limitations

Several limitations are present in the current project.

1. The corpus was selected for open data, that were successfully computationally reproducible and relatively strong target results, so the observed concordance rate is likely optimistic and is not representative of the population.

2. Some papers contribute more than one target, so the 47 observations are not fully independent.
3. Bayes factors from different statistical families are not generated by one universal prior construction. Some rely on established default priors, some on fractional Bayes factors, and some on project-calibrated priors. Consequently, cross-claim comparisons of exact BF magnitude should be made cautiously.
4. The $BF_{10} = 3$ and $1/3$ boundaries are conventional reporting thresholds. Claims close to those values should be interpreted continuously and with prior sensitivity rather than as qualitatively different merely because they fall on opposite sides of the threshold.
5. The project chose to keep reanalysis as faithful to the original analysis as possible. Therefore, with the exception of two targets, the project did not independently research whether every source analysis used the scientifically ideal test or hypothesis, since our primary question was what happened when those reproduced targets were reanalysed under Bayesian evidence.

5.6 Further Research

The most direct extension is to apply the same workflow to a broader and less favourably selected corpus, including closed-data studies where data later become available, non-primary target results, and other areas of psychology. This would test whether the high concordance and low inconclusive rates observed persists when weaker and less reproducibility-friendly results are included.

A second extension would be a blinded many-analyst reanalysis. Independent analysts could be given the same target claims and underlying data, but not the reported frequentist results or the Bayesian analyses developed here, and asked to construct what they regard as the most defensible Bayesian analysis. This would make it possible to study how much variation arises from reasonable differences in analyst judgement, including decisions about exclusions and outliers, missing-data handling, variable construction, transformations, as well as choices in likelihood, hierarchical structure, sidedness, and prior specification. Consequently, it would highlight how a Bayesian reanalysis that tries to stay as faithful as possible to the original study differs from a Bayesian analysis that was built from scratch.

Thirdly, a multiverse analysis could be implemented in which each target claim is evaluated across a prespecified set of defensible Bayesian formulations. This would show if the obtained Bayes factors would be robust across reasonable analytic choices.

Finally, wave 3 points to a possible methodological gaps: established Bayesian analogues exist for several common rank procedures, but could not be found for the omnibus repeated-measures rank test (such as Study 45). Further work on validated Bayesian rank-based repeated-measures testing could reduce the need for claim-level reconstruction in future re-analyses.

6. Conclusion

This project was conducted as part of the Bristol Summer Research Internship in STEM under the supervision of Dr Peter Allen and Dr Robbie Clark, who proposed the research topic and provided guidance throughout the project. The Bayesian reanalysis builds on earlier computational reproduction work by Ali Almuhanha, whose reproduced frequentist results formed the empirical starting point for the present analyses.

We asked if once a published statistical result could be recovered from the original data, how strong was the evidence for the scientific claim under a Bayesian analysis?

Across 47 computationally reproduced psychology-of-teaching results, the Bayesian reanalysis produced no reversals in concordance: 39 targets were concordant, 8 were inconclusive, and none were discordant. The contribution of the Bayesian analysis was not to overturn conclusions but to grade them, by separating decisive from borderline results, and, among non-significant findings, separating genuine evidence of absence from mere absence of evidence, a distinction the frequentist dichotomy cannot express.

The broader methodological contribution was the correspondence-first protocol and the heterogeneous analysis. On the one hand, Bayesian reanalysis preserved the reproduced statistical hypothesis when a defensible counterpart existed, which allowed for a more faithful reproduction, and when one could not be found, then a Bayesian model that directly represented the inherited claim and original data structure was computed. On the other hand a strength of this project is the variety of statistical analyses it includes. Some previous large-scale Bayesian reanalyses have focused on a single analysis family, our corpus instead includes *t*-tests, correlations, ANOVAs, heteroscedastic models, multivariate tests, rank-based analyses, mixed-effects models, structural-equation models, and longitudinal and ordinal models. This makes the project less statistically uniform, but also closer to the range of methods used in published research.

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A. Appendix A: Detailed data tables

Table A.1: Frequentist results for inconclusive cases

Claim ID*	Original model	Frequentist test	Sample size	Frequentist result**
S20C01	Planned linear contrast within a latent difference score SEM	sem_contrast	$N = 153$	Estimate = .372, $SE = .183$, $p = .021$
S30C01	Paired samples t -test	paired_t_test	$N = 37$	$t(36) = -1.21$, $p = .233$
S37C02	Mixed-effects binomial logistic regression	mixed_effects_logistic_regression	$N = 563$	log-odds = -1.11 , $p = .007$
S43C01	$2 \times 2 \times 2$ mixed ANOVA	mixed_anova	$N = 67$	$F(1, 64) = 5.33$, $\eta_p^2 = .08$, $p = .024$
S47C01	2×3 mixed ANOVA	independent_t_test	$n_1 = 63$, $n_2 = 58$, $N = 121$	$F(1, 119) = 4.98$, $p = .027$
S47C02	2×3 mixed ANOVA	mixed_anova	$n_1 = 63$, $n_2 = 58$, $N = 121$	$F(2, 238) = 2.90$, $\eta_p^2 = .024$, $p = .057$
S49C01	Paired samples t -test	paired_t_test	$N = 28$	$t(27) = 1.20$, $d = .23$, $p = .119$
S60C01	2×2 between-subjects factorial ANOVA	factorial_between_anova	$n_1 = 163$, $n_2 = 163$, $N = 326$	$F(1, 322) = 5.17$, $\eta_p^2 = .02$, $p = .024$

Note.

* Claim IDs are abbreviated as $SxCy$, where S denotes the study number and C denotes the claim number.

** Exact p -values are taken from the reproduction results and rounded to three decimal places. S37C02 is an exception: its reproduced p -value differs from the originally reported value because the standard error of the mixed-effects logistic regression is sensitive to the `lme4` version used; however, both results lead to the same substantive conclusion.

Table A.2: Bayesian results for inconclusive cases

Claim ID	Bayesian model	Null model	Alternative model	BF_{10}^*	Concordance
S20C01	Adjusted fractional order-restricted	Test_H1a = 0	Test_H1a > 0	1.61	Significant–inconclusive
S30C01	t -test	$\delta = 0$	$\delta \neq 0$	0.35	Non-significant–inconclusive
S37C02	Full Bayesian GLMM	Full Bayesian GLMM without synchronous11	Full Bayesian GLMM with synchronous11	2.95	Significant–inconclusive
S43C01	Gaussian adjusted contrast	outcome_z ~ 0 + nuisance_1 + nuisance_2	outcome_z ~ 0 + focal + nuisance_1 + nuisance_2	2.34	Significant–inconclusive
S47C01	t -test	$\delta = 0$	$\delta \neq 0$	1.80	Significant–inconclusive
S47C02	ANOVA Cauchy	score ~ Condition + format + participant	score ~ Condition + format + Condition format + participant	0.69	Non-significant–inconclusive
S49C01	t -test	$\delta = 0$	$\delta > 0$	0.67	Non-significant–inconclusive
S60C01	ANOVA Cauchy	outcome ~ B + A! :!B	outcome ~ A + B + A! :!B	1.28	Significant–inconclusive

Note.

* All reported Bayes factors are based on the primary prior specification.

δ denotes the standardized effect parameter in the Bayesian t -test.

There is heterogeneity within the inconclusive range. Although all eight targeted claims are classified as inconclusive in the Bayesian reanalysis, their Bayes factors vary significantly. Some are very close to the evidential threshold: S30C01 and S37C02 have their $BF_{10} = 0.35$ and $BF_{10} = 2.95$ correspondingly, which are close to the evidential threshold of $1/3$ and 3 for favoring the null/alternative model. In contrast, some are more centrally located, such as S60C01 with its $BF_{10} = 1.28$, which provides very limited evidence for differentiating between the two models.

Inconclusive evidence is observed across a wide range of sample sizes, from $N = 37$ to $N = 563$, and is not only confined, as intuitively inferred, to the small sampled target. The claims with relatively large sample sizes, such as S60C01 ($N = 326$, $BF_{10} = 1.28$) and S37C02 ($N = 563$, $BF_{10} = 2.95$), remain inconclusive.

Table A.3: Prior scale sensitivity for inconclusive cases

Claim ID	Bayesian model	Sensitivity specification 1	Primary	Sensitivity specification 2
S20C01	Adjusted fractional order-restricted	sensitivity b_2 , $BF_{10} = 2.28$	primary, $BF_{10} = 1.61$	sensitivity b_3 , $BF_{10} = 2.79$
S30C01	t -test	$r = .500$, $BF_{10} = 0.45$	$r = .707$, $BF_{10} = 0.35$	$r = 1.000$, $BF_{10} = 0.26$
S37C02	Full Bayesian GLMM	$r = .300$, $BF_{10} = 1.64$	$r = .600$, $BF_{10} = 2.95$	$r = 1.000$, $BF_{10} = 3.70$
S43C01	Gaussian adjusted contrast	$r = .300$, $BF_{10} = 3.43$	$r = .600$, $BF_{10} = 2.34$	$r = 1.000$, $BF_{10} = 1.52$
S47C01	t -test	$r = .500$, $BF_{10} = 2.10$	$r = .707$, $BF_{10} = 1.80$	$r = 1.000$, $BF_{10} = 1.44$
S47C02	ANOVA Cauchy	$r = .250$, $BF_{10} = 1.58$	$r = .500$, $BF_{10} = 0.69$	$r = .707$, $BF_{10} = 0.43$
S49C01	t -test	$r = .500$, $BF_{10} = 0.85$	$r = .707$, $BF_{10} = 0.67$	$r = 1.000$, $BF_{10} = 0.51$
S60C01	ANOVA Cauchy	$r = .250$, $BF_{10} = 2.24$	$r = .500$, $BF_{10} = 1.28$	$r = .707$, $BF_{10} = 1.01$

For S20C01, the sensitivity specifications are labelled *sensitivity* b_2 and *sensitivity* b_3 rather than being parameterized by r .

B. Appendix B: Dataset figures

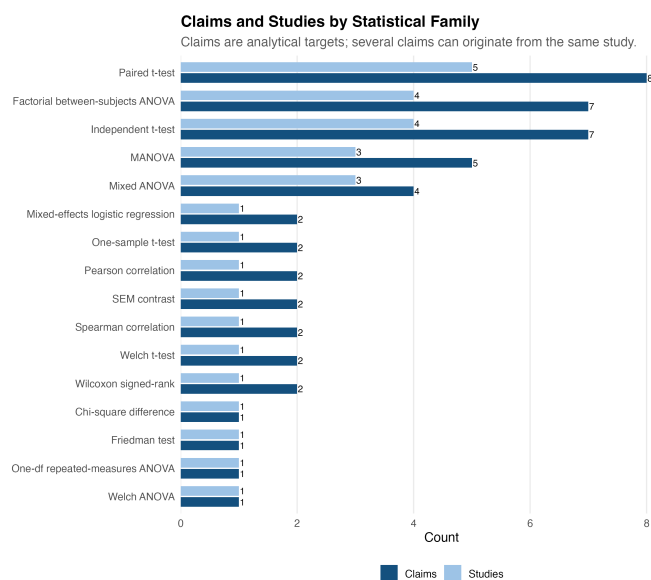


Figure B.1: Number of claims and studies by statistical family

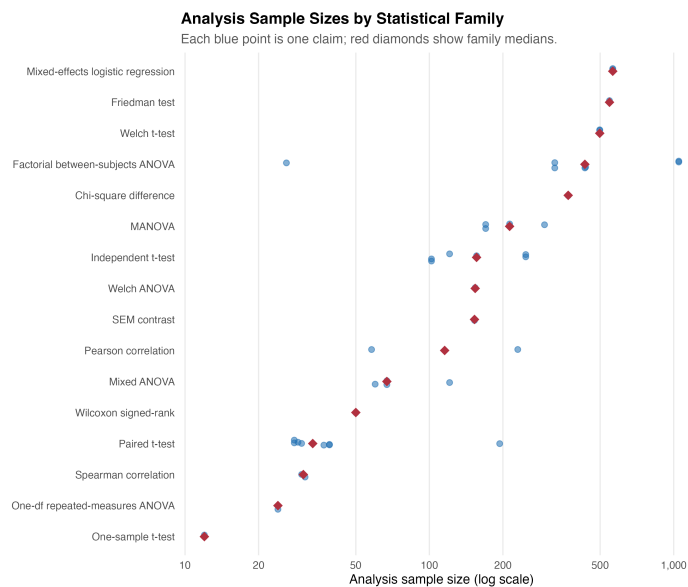


Figure B.2: Sample size by family

C. Appendix C: Raw Data Manifest

Table C.1: Original data sources for studies included in the Bayesian reanalysis.
The files stored [\[here\]](#) in the repository. The manifest can also be found [\[here\]](#).

Study	Authors	Article	Open data
03	Virginia B. Wickline; Carmen Ford; Regan A. R. Gurung; Drew Appleby	<i>Can Syllabus Snapshots Predict Student Perceptions of Course and Instructor?</i>	OSF: gtyaj
05	Bridgette M. Hard; Joshua M. Lovett; Shannon T. Brady	<i>What Do Students Remember About Introductory Psychology, Years Later?</i>	OSF: tvgky
06	Alison E. Kelly; Virginia Clinton-Lisell	<i>Strengthening Online Psychology Students' Sense of Belonging with Social Annotation: An Experimental Study</i>	OSF: zcqjr
10	Alyssa C. Smith; David C. S. Filice; Heather Poole; Ayesha Khan; Kate Whalen; Daniel Smilek	<i>Indicators of Student Well-Being in Canadian Undergraduates Before and During the COVID-19 Pandemic</i>	OSF: j8qph
13	Carlee Beth Hawkins; Alexis Z. Camp; Matthew P. Schunke	<i>Experimental Evaluation of Implicit Bias Education in the College Classroom</i>	OSF: my6dj
14	Stephanie M. Anglin; John E. Edlund	<i>Perceived Need for Reform in Field-Wide Methods and the Teaching of Replication, Interpretation, and Transparency</i>	OSF: g5qdz
18	Madeleine Pownall; Chloe Thompson; Pam Blundell-Birtill; Samantha J. Newell; Richard Harris	<i>Does "Psychological Literacy" Feature in Non-Psychology Degrees? A Cross-Discipline Study of Student Perceptions</i>	OSF: 9f5mb
20	Tom Rosman; Martin Kerwer	<i>Mechanisms of Epistemic Change: The Roles of Reflection and Social Interaction</i>	PsychArchives
21	William J. Chopik; Ryan H. Bremner; Andrew M. Defever; Victor N. Keller	<i>How (and Whether) to Teach Undergraduates About the Replication Crisis in Psychological Science</i>	OSF: mh9pe
22	Nina Udvardi-Lakos; Marlene Weirich; Kim Lützenburger; Julia Asbrand; Alexander Renkl	<i>Combining a Direct and Indirect Training Approach for Cross-Domain Competences: The Case of the Course "Pedagogy for Psychotherapists"</i>	OSF: dzpxs
26	Alison E. Kelly; Jenna N. Laurin; Virginia Clinton-Lisell	<i>Making Psychology's Hidden Figures Visible Using Open Educational Resources: A Replication and Extension Study</i>	OSF: rnpz2
27	William J. Chopik; Jeewon Oh	<i>Implementing the Fast Friends Procedure to Build Camaraderie in a Remote Synchronous Teaching Setting</i>	OSF: rvcmy
29	Beth Morling; Jeong Min Lee	<i>Are "Associate Professors" Better Than "Associate Teaching Professors"? Student and Faculty Perceptions of Faculty Titles</i>	OSF: ptvg4
30	Alan Jern	<i>A Preliminary Study of the Educational Benefits of Conducting Replications in the Classroom</i>	OSF: 7jj5t

Continued on next page

Table C.1 – continued

Study	Authors	Article	Open data
33	Gili Freedman; India G. Oates; Shannon Elise Kirk	<i>Applying Social Psychology to a Global Crisis: Student Engagement in a Laboratory Class During the COVID-19 Pandemic</i>	OSF: k4vcy
35	Roberta Ekuni; Ohana T. Macacare; Sabine Pompeia	<i>Reducing the Negative Effects of Multitasking on Online or Distance Learning by Using Retrieval Practice</i>	OSF: ymezx
37	C. Donnan Gravelle; Raoul Roberts; Elizabeth S. Che; Arshia K. Lodhi; Nicole M. Zapparrata; Teresa M. Ober; Jessica E. Brodsky; Patricia J. Brooks	<i>Online Course Formats and Student Self-Efficacy in Academic Skills Predict Persistence in Introductory Psychology</i>	OSF: wabvs
39	Ashley M. Biddle; Virginia Clinton-Lisell	<i>“The Pictures Allowed Me to Connect to the Material More”: Student Perceptions of a Diversity-Focused Open Pedagogy Assignment</i>	OSF: wdu5k
40	Sydney Y. Wood; Victoria L. Cross	<i>Improving Students’ Understanding Through Metacognition About Instructor Feedback: A Causal Modeling Approach</i>	OSF: a6gbn
41	Andrew J. Kelly; Audrey E. Parrish	<i>Don’t Pop It! Modifying the Balloon Analogue Risk Task for a Course-Based Research Experience</i>	OSF: 7naj9
43	Julia Glaser; Tobias Richter	<i>The Testing Effect in the Lecture Hall: Does It Transfer to Content Studied but Not Practiced?</i>	OSF: wc3kh
44	Michelle L. Rivers; Addison L. Babineau; Katherine P. Neely; Sarah K. Tauber	<i>How Do Students and Faculty Consider Numerical Ratings and Comments About Daily Quizzing When Interpreting Student Evaluations of Teaching?</i>	OSF: qxs2m
45	Adam John Privitera; Jue Wang; Xin Jiang	<i>Understanding of Psychological Literacy: Comparative Insights From Undergraduates in China and the United Kingdom</i>	OSF: z5xj8
47	Guy A. Boysen; Peyton N. Osgood	<i>Test Format, Learning Confidence, and Perceptions of Teaching Effectiveness</i>	OSF: 7txp2
49	Leamarie T. Gordon; Gregory I. Hughes; Amy M. Smith	<i>Modernizing the Flipped Classroom: Replacing Lecture Time With Asynchronous Spaced Retrieval</i>	OSF: 54qgs
51	Virginia Clinton-Lisell; Alison E. Kelly	<i>Are Scientific Memes Motivating and Does Public Sharing Affect Motivation?</i>	OSF: 96fbr
53	Michelle Wong; Leigh M. Marshall; Haley C. Blank; Bridgette Martin Hard	<i>Up Close and Personal: Examining Effects of Instructor Video Presence on Students’ Sense of Connection</i>	OSF: rhzc8
55	Lauren E. Bates; Sarah J. Myers; Edward L. DeLosh; Matthew G. Rhodes	<i>The Learning Benefits of Mastery Quizzing Versus Other Quizzing Methods</i>	OSF: kw9n7
60	Melanie R. Maimon; Kristina Howansky; Diana T. Sanchez	<i>Fostering Inclusivity: Exploring the Impact of Identity Safety Cues and Instructor Gender on Students’ Impressions and Belonging</i>	OSF: u2rnp

D. Appendix D: Pipeline Design

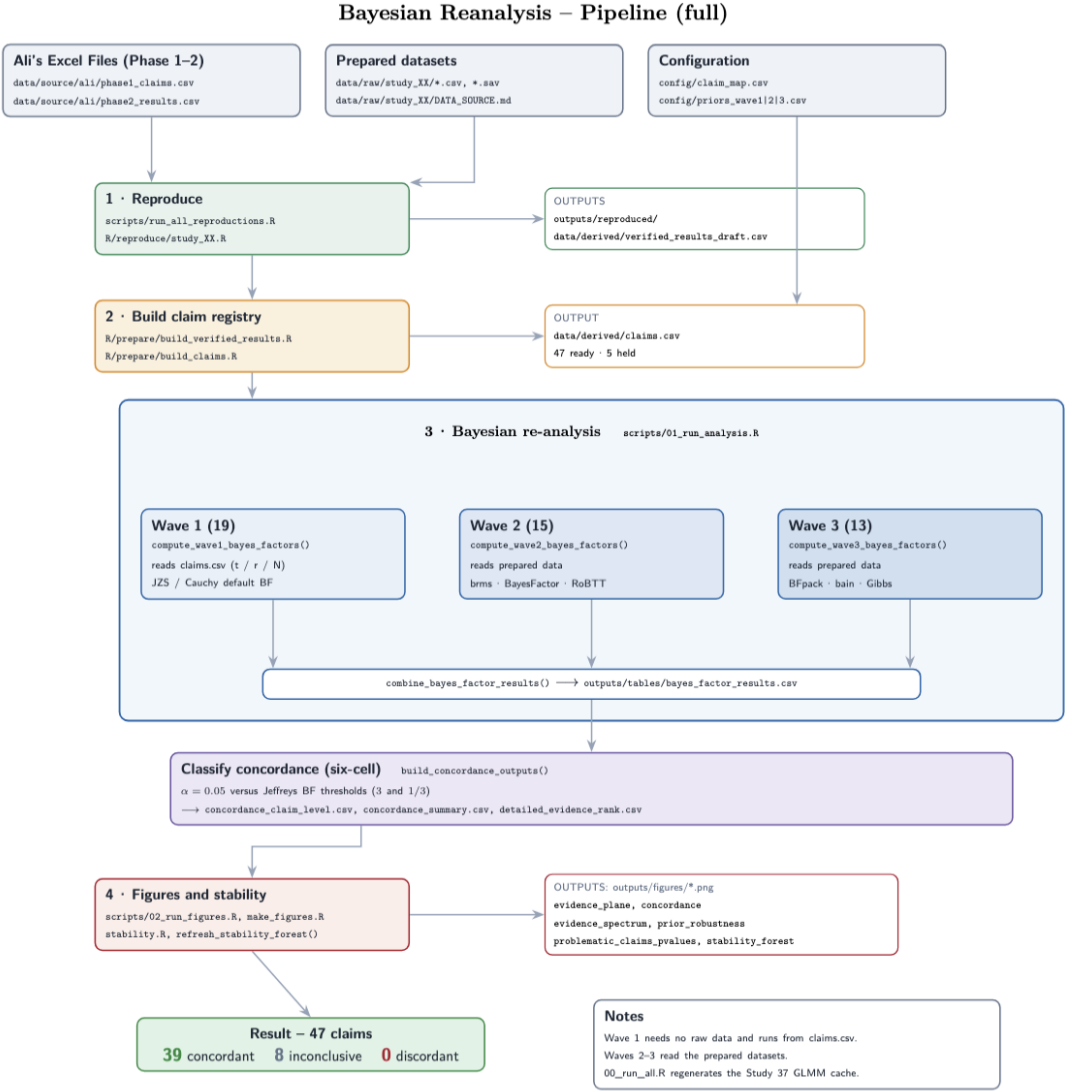


Figure D.1: Detailed schema of the pipeline

E. Appendix E: Numerical and Algorithmic Stability

E.1 Numerical and Algorithmic Stability

Because several Bayes factors were estimated using Monte Carlo, MCMC, or bridge-sampling procedures, numerical stability was assessed separately from prior sensitivity.

When applicable, the primary estimate was compared with its estimated numerical range. Depending on the model family, this range was obtained from repeated bridge-sampling estimates, Monte Carlo error, repeated Gibbs runs under different random seeds, or MCMC convergence diagnostics. Analytic Bayes factors have no corresponding Monte Carlo error and are shown as exact.

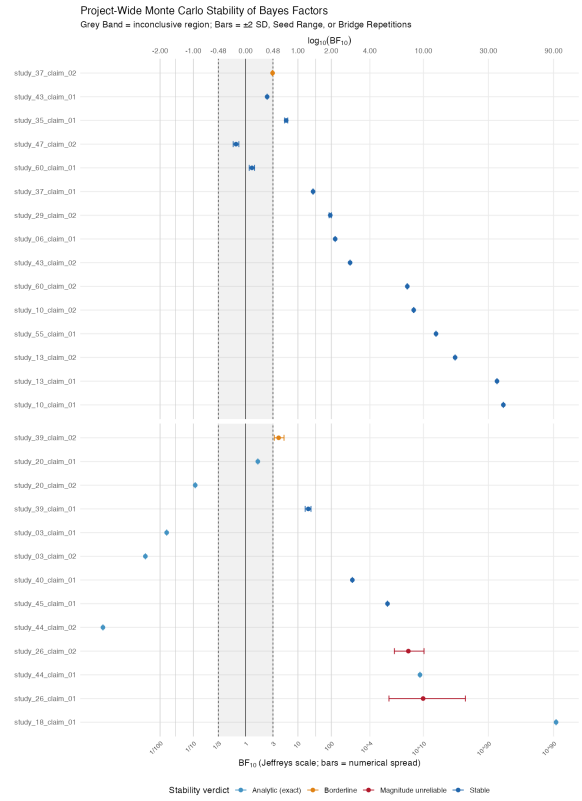


Figure E.1: Algorithmic and Numerical Stability Diagnostics (better visualisation [\[here\]](#))

Figure E.1 displays the numerical range of each Bayes-factor estimate relative to the evidence thresholds $BF_{10} = 1/3$ and $BF_{10} = 3$. Numerical instability was generally too small to alter the substantive evidence classification. Study 37 Claim 2 was numerically close to the $BF_{10} = 3$ boundary and was therefore flagged as borderline, although its estimated numerical range remained within the inconclusive category. Cases flagged as numerically unstable (see Figure E.1 in red) were exclusively cases where evidence was overwhelming concordant and instability can likely be attributed precisely to those extreme values. The remaining stochastic estimates were either classification-stable or analytically exact.